BTEUP EXAM NOTES | ENGLISH EDITION

Information Security

Unit 1: Introduction, PAIN, OS Security, RAID and Anti-Virus

About these notes

Part A explains every topic in very simple English with real-life examples, so a beginner can understand it from zero.
Part B gives important exam questions with full 5-mark model answers, ready to write in the exam.
At the end: quick revision sheet, common mistakes, and a self-test.

Syllabus covered (Unit 1)

Introduction to Information Security, Various aspects of information security (PAIN), Security Features of Operating Systems: Authentication, Logs, Audit Features, File System Protection, User Privileges, RAID options, Anti-Virus Software.

Unit Overview and Contents

Simple idea: Information Security means protecting data from the wrong people, and making sure the right people can use it at the right time. This unit covers 3 things: (1) the basics and PAIN, (2) security features inside an Operating System, and (3) extra protection using RAID and Anti-virus.

Topic	Exam importance	Where in notes
Information Security, CIA Triad	High	Part A: Topic 1 | Part B: Q1, Q9
PAIN (4 aspects)	Very High (almost sure)	Part A: Topic 2 | Part B: Q1, Q12
OS Security: Authentication, Logs, File protection, Privileges	Very High	Part A: Topic 3 | Part B: Q3 to Q8
RAID options	Very High	Part A: Topic 4 | Part B: Q2, Q10
Anti-Virus Software	Very High	Part A: Topic 5 | Part B: Q4, Q11

Contents

Part A: Easy Explanation	Part B: Important Questions with Answers
Topic 1: Information Security and CIA Triad Topic 2: PAIN Topic 3: OS Security Features Topic 4: RAID Topic 5: Anti-Virus Software	Q1 Information Security and PAIN (Very Likely) Q2 RAID and its levels (Very Likely) Q3 OS security features (Very Likely) Q4 Anti-virus software (Very Likely) Q5 Authentication (Likely) Q6 Logs and Audit (Likely) Q7 File system protection (Likely) Q8 User privileges (Likely) Q9 CIA Triad (Likely) Q10 RAID 0 vs RAID 1 (Possible) Q11 Signature vs Heuristic (Possible) Q12 Non-repudiation (Possible)
Then: Quick Revision Sheet, Common Mistakes, Self-Test with answers	

How to study this unit (suggested time: 1 day)

1) Read Part A once to understand. 2) Read Part B answers and copy each answer on paper with the diagram. 3) Revise using the Quick Revision Sheet. 4) Attempt the Self-Test without looking.

PART A: Easy Explanation (Simple English)

Topic 1: Information Security and CIA Triad

Real-life example

Imagine you keep money in a cupboard at home. Putting a **lock** on it is security. Information Security puts a lock on data in the same way, so that no wrong person can see, change or delete it.

Definition

Information Security (InfoSec) is the practice of protecting information from unauthorized access, use, disclosure, modification or destruction.

Key points

- It is not only about computer data. It protects **every kind of information** (paper, digital, spoken).
- It has **3 main goals**, called the **CIA Triad**.
- **Confidentiality:** only authorized people can see the data. Example: your bank password is known only to you.
- **Integrity:** data is not changed without permission. Example: nobody can edit the marks on your marksheet.
- **Availability:** data is available whenever needed. Example: the bank server runs 24x7.

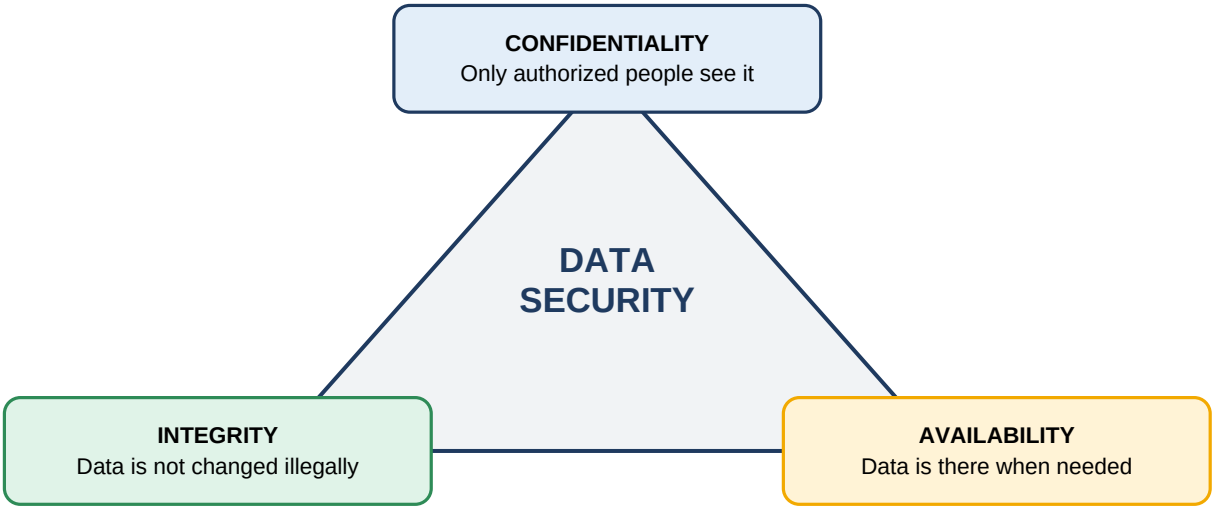


Figure 1: CIA Triad

Topic 2: PAIN (Aspects of Information Security)

Real-life example

Suppose you make an online payment. The system must make sure four things: nobody else can see your data (**Privacy**), it is really you making the payment (**Authentication**), the amount was not changed on the way (**Integrity**), and later you cannot say "I did not do it" (**Non-repudiation**).

Definition

PAIN stands for **P**rivacy, **A**uthentication, **I**ntegrity and **N**on-repudiation. These are the four main aspects (pillars) of information security.

Aspect	Meaning	How it is achieved	Example
Privacy	Only authorized persons can see the data	Encryption, access control	WhatsApp end-to-end encryption
Authentication	Verifying the identity of the user	Password, OTP, biometrics	ATM PIN
Integrity	Data is not changed in transit or storage	Hash, checksum, digital signature	Matching the hash after downloading a file
Non-repudiation	Sender cannot deny later	Digital signature	Cannot deny an online payment afterwards

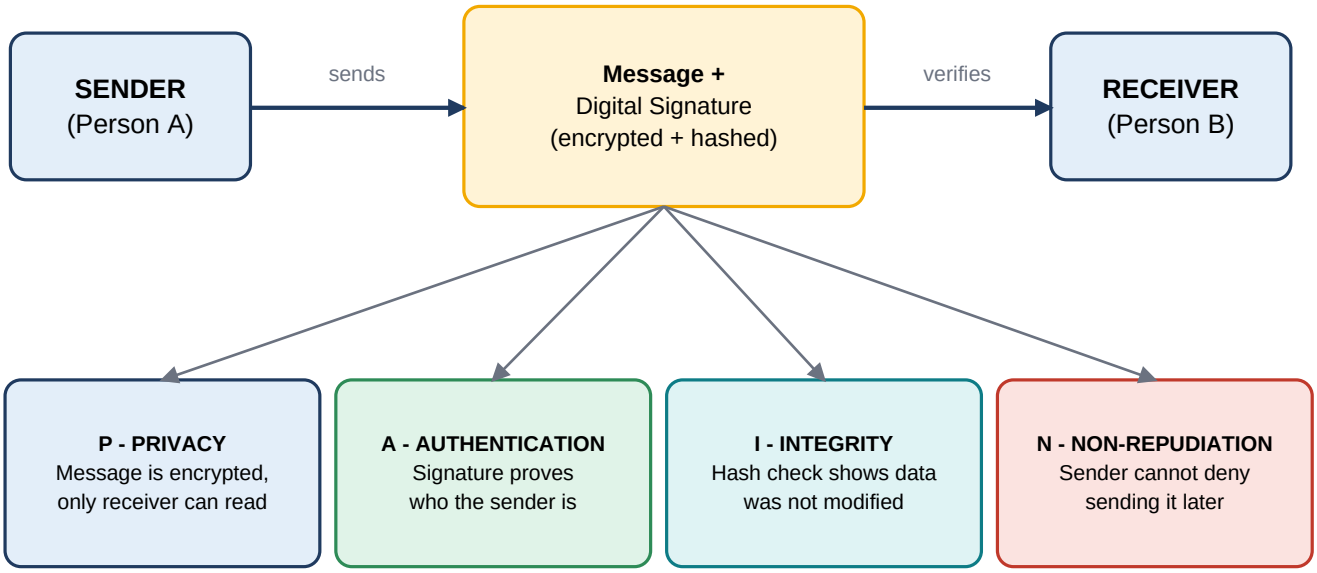


Figure 2: PAIN with Digital Signature flow

Remember

Privacy is also called **Confidentiality**. Authentication and **Authorization** are different: Authentication = "Who are you?", Authorization = "What are you allowed to do?"

Topic 3: Security Features of Operating Systems

An Operating System (Windows, Linux, Android) is the manager of the computer. So its security is the first line of defence. An OS has **4 main security features**:

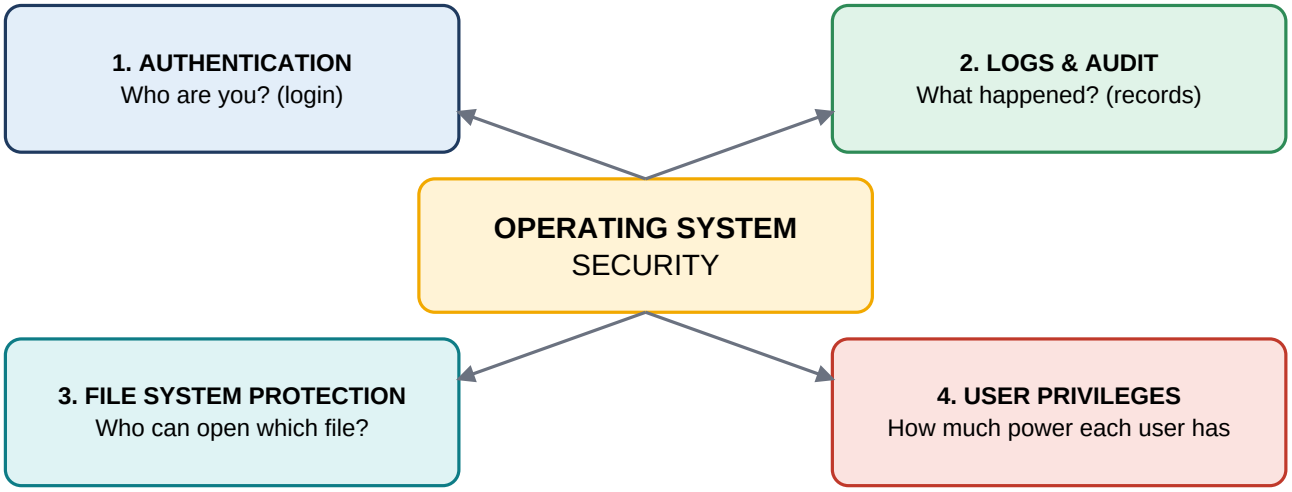


Figure 3: OS Security Features

3(a) Authentication

> **Real-life example**
> When you withdraw money at a bank, the staff asks for an **ID proof**. A computer also asks for proof of identity at login.

Definition: Authentication is the process of verifying the identity of a user before giving access to the system.

- **Something you know:** Password, PIN.
- **Something you have:** OTP, smart card, token.
- **Something you are:** Fingerprint, face, iris (biometrics).
- Using 2 or more factors together is called **Multi-Factor Authentication (MFA)**. Example: ATM card + PIN.

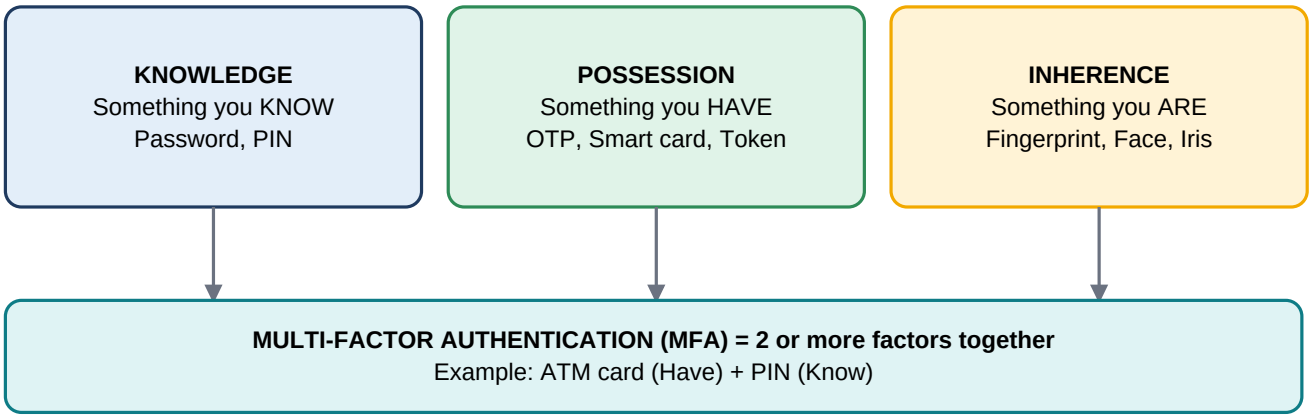


Figure 4: Authentication factors

3(b) Logs and Audit Features

> **Real-life example**
> Just like a CCTV camera records everything in a shop, **logs** record everything that happens in a computer.

Log: A record of events happening in the system (who logged in, when, what was changed). **Audit:** Examining these logs to find suspicious or illegal activity.

- Failed login attempts, file access, program errors and system changes are recorded.
- After a hacking attempt, logs show **what happened, when it happened and who did it**.
- **Windows:** Event Viewer. **Linux:** `/var/log/` folder.
- Types: **Security log, System log, Application log**.

3(c) File System Protection

> **Real-life example**
> Every room in a house can have a different key. In the same way, every file or folder can have different permissions.

Definition: File system protection controls who can read, write or execute a file or folder.

- **Permissions:** Read (r), Write (w), Execute (x).
- **Linux:** permissions are set separately for Owner, Group and Others. Command: `chmod`.
- **Windows (NTFS):** an **ACL (Access Control List)** decides who can do what.
- **Encryption: EFS** (file level) and **BitLocker** (full disk) lock data in Windows.

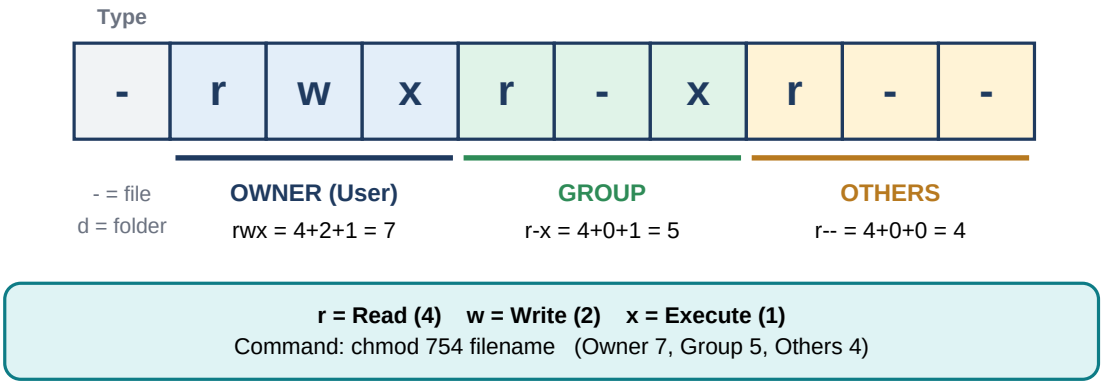


Figure 5: Linux file permission -rwxr-xr--

3(d) User Privileges

> **Real-life example**
> In an office, the manager has keys to all cabinets, but the peon has keys to only a few. This reduces loss from mistakes or theft.

Definition: User privileges are the rights and permissions given to a user account to perform actions on the system.

- **Principle of Least Privilege:** give a user only the rights needed for the job.
- **Administrator / root:** full control. **Standard user:** limited control. **Guest:** very little access.
- **Windows UAC** (User Account Control): asks for permission before a system change.
- **Linux sudo:** gives temporary admin power.
- Benefit: less damage from malware or accidental mistakes.

Topic 4: RAID Options

Real-life example

If you have only one notebook and you lose it, all your notes are gone. If you write the **same notes in 2 notebooks**, you are safe even if one is lost. RAID joins **many hard disks** in the same way to make data safe or fast.

Definition

RAID (Redundant Array of Independent Disks) is a technique that combines multiple physical disks into one logical unit to improve performance and/or data protection.

3 basic techniques

- **Striping:** splitting data into pieces across disks. Speed increases.
- **Mirroring:** keeping an identical copy on another disk. Safety increases.
- **Parity:** extra calculated data used to rebuild the data of a failed disk.

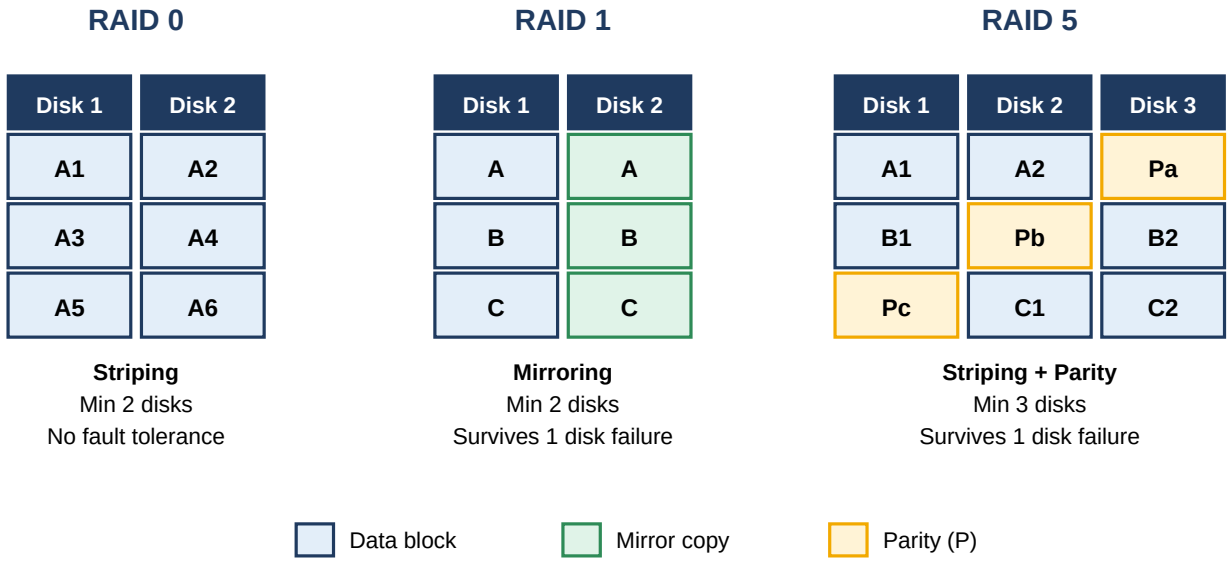


Figure 6: RAID 0, RAID 1 and RAID 5 layouts

RAID Level	Technique	Min disks	Fault tolerance	Advantage / Disadvantage
RAID 0	Striping	2	None (1 disk fail = all data lost)	Very fast, but not safe
RAID 1	Mirroring	2	1 disk can fail	Safe, but 50% space wasted
RAID 5	Striping + distributed parity	3	1 disk can fail	Good balance of speed and safety
RAID 6	Striping + double parity	4	2 disks can fail	Safer, slightly slower
RAID 10	Mirroring + striping	4	1 disk per mirror pair	Fast and safe, but expensive

Very important

RAID is not a replacement for backup. If a file is deleted or a virus enters, RAID does not help, because the wrong data is copied to all disks as well. Keep a separate backup.

Topic 5: Anti-Virus Software

Real-life example

Anti-virus is the **security guard** of your computer. Like a guard checks everyone at the gate, anti-virus checks every file.

Definition

Anti-virus software is a program that detects, blocks and removes malware such as viruses, worms and trojans.

Detection methods (how it works)

- **Signature-based:** matches files against a database of known virus signatures. Cannot catch new viruses.
- **Heuristic-based:** guesses by looking at suspicious code or style of a program.
- **Behaviour-based:** watches what a program does while running (for example, encrypting files).
- **Sandboxing:** runs a suspicious file in a separate safe area to test it.

Main features

- **Real-time protection:** always scanning in the background.
- **Full / scheduled scan:** checks the whole system.
- **Quarantine:** keeps an infected file isolated in a safe folder.
- **Auto-update:** virus definitions are updated regularly.
- Email and web protection. Examples: Windows Defender, Quick Heal, Kaspersky, Norton.

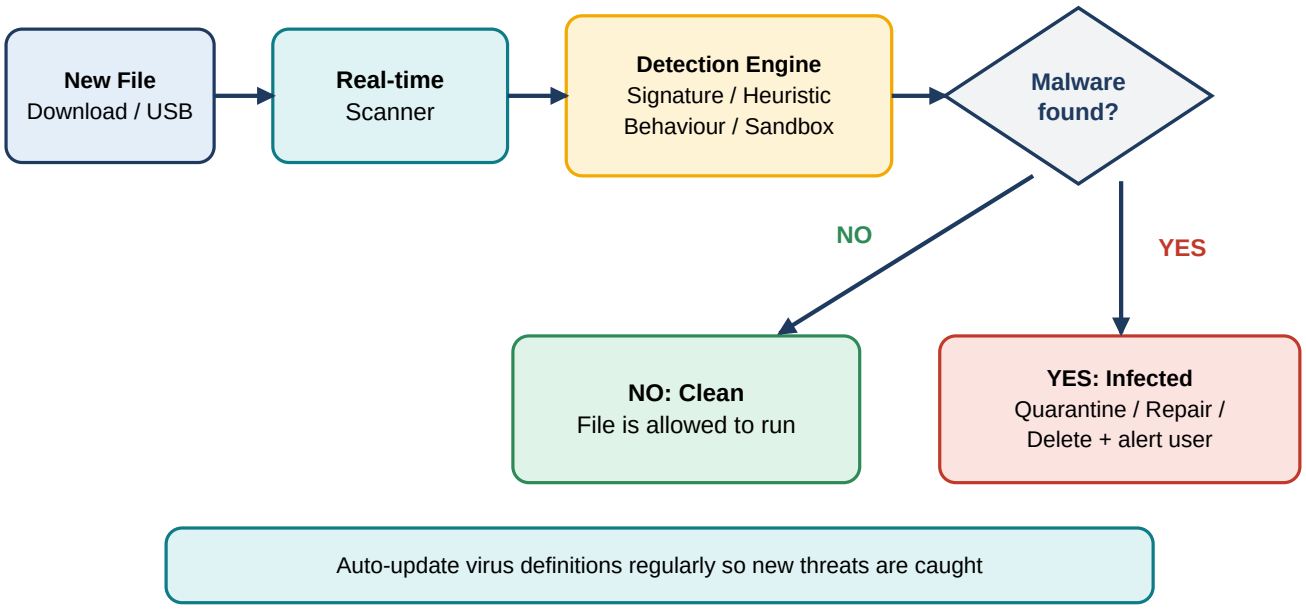


Figure 7: Working of Anti-virus software

Limitation

Zero-day (brand new) viruses may not be caught at first, so regular updates are necessary.

PART B: Important Questions with Model Answers (English)

Priority: **Very Likely** = appears most often in BTEUP. **Likely** = good chance. **Possible** = can appear. Each answer follows the exam format: **Definition, Points, Diagram/Table, Conclusion**.

Q1. What is Information Security? Explain the PAIN aspects. — VERY LIKELY

Definition / Introduction: Information Security is the practice of protecting information from unauthorized access, use, disclosure, modification or destruction. Its main aspects are known as **PAIN**: Privacy, Authentication, Integrity and Non-repudiation.

Points

- **Privacy:** information is accessible only to authorized persons. It is achieved by encryption and access control.
- **Authentication:** verifying that a user or sender is genuinely who they claim to be, using passwords, OTP or biometrics.
- **Integrity:** information is not altered during storage or transmission. It is ensured by hash functions, checksums and digital signatures.
- **Non-repudiation:** the sender cannot later deny having sent a message or made a transaction. It is provided by digital signatures.

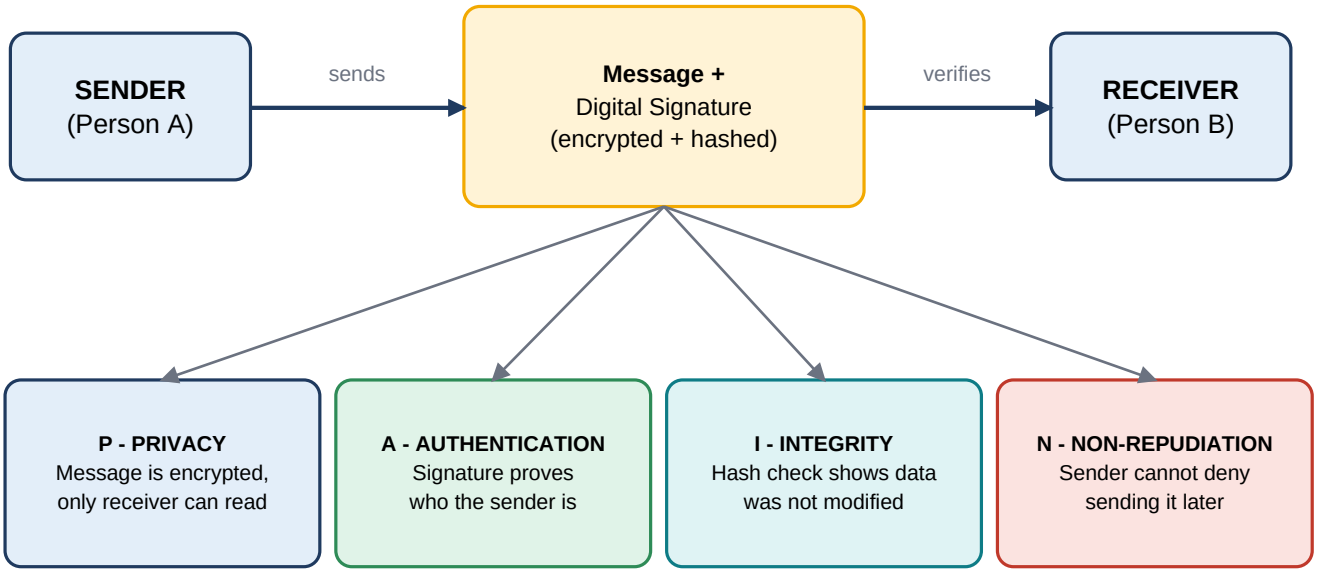


Figure: PAIN with digital signature

Conclusion: PAIN together ensures that information stays secret, genuine, unchanged and legally accountable.

Q2. What is RAID? Explain RAID levels 0, 1 and 5. VERY LIKELY

Definition / Introduction: RAID (Redundant Array of Independent Disks) is a technology that combines multiple physical disks into a single logical unit to improve performance, fault tolerance, or both.

Points

- **Striping** splits data across disks for speed, **mirroring** keeps identical copies, and **parity** stores extra data to rebuild a failed disk.
- **RAID 0 (Striping):** minimum 2 disks. Very fast read/write, but no fault tolerance. Failure of one disk loses all data.
- **RAID 1 (Mirroring):** minimum 2 disks. Same data on both disks. Survives one disk failure, but only 50% of the capacity is usable.
- **RAID 5 (Striping with distributed parity):** minimum 3 disks. Survives one disk failure. Good balance of speed, safety and storage use.
- RAID 6 uses double parity (min 4 disks, survives 2 failures). RAID 10 combines mirroring and striping (min 4 disks).

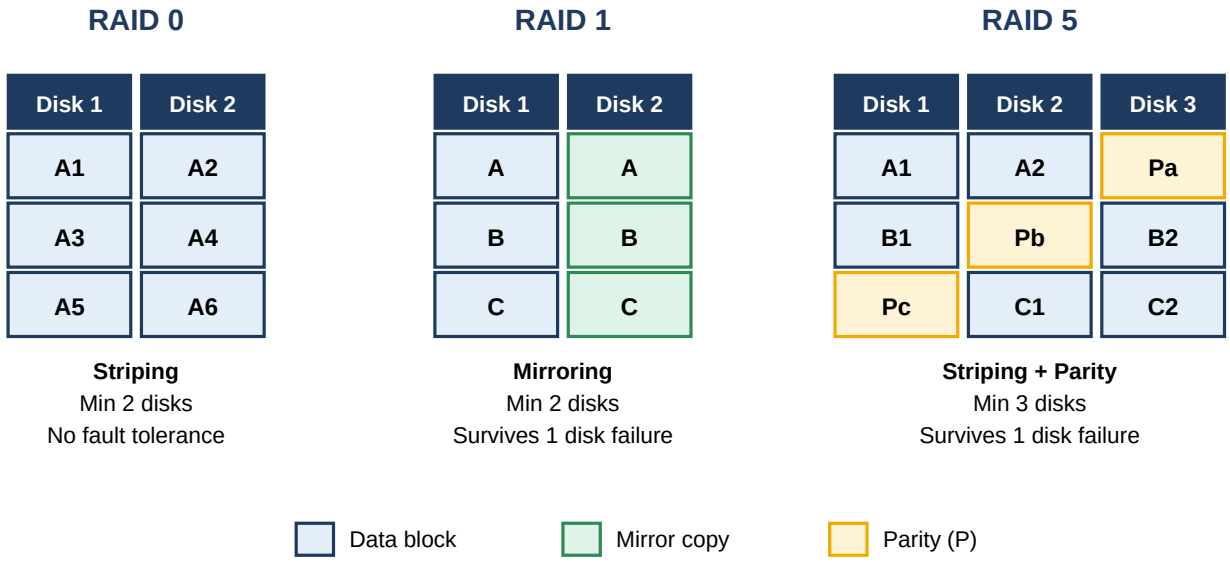


Figure: RAID 0, 1 and 5

Level	Min disks	Fault tolerance	Speed	Space use
RAID 0	2	None	Very high	100%
RAID 1	2	1 disk	Good read	50%
RAID 5	3	1 disk	High	(n-1)/n

Conclusion: RAID improves speed and availability of data, but it is not a replacement for regular backups.

Q3. Explain the security features of an Operating System. VERY LIKELY

Definition / Introduction: Operating System security means protecting system resources (files, memory, devices) from unauthorized access and misuse. Modern operating systems provide the following features.

Points

- **Authentication:** verifies user identity using password, PIN, smart card or biometrics before allowing login.
- **Logs and Audit:** record system events such as logins, file access and errors, so suspicious activity can be traced.
- **File System Protection:** permissions (read, write, execute), ACLs and encryption (EFS, BitLocker) control access to files.
- **User Privileges:** each user gets only the rights needed for the job (least privilege). Examples: Windows UAC, Linux sudo.

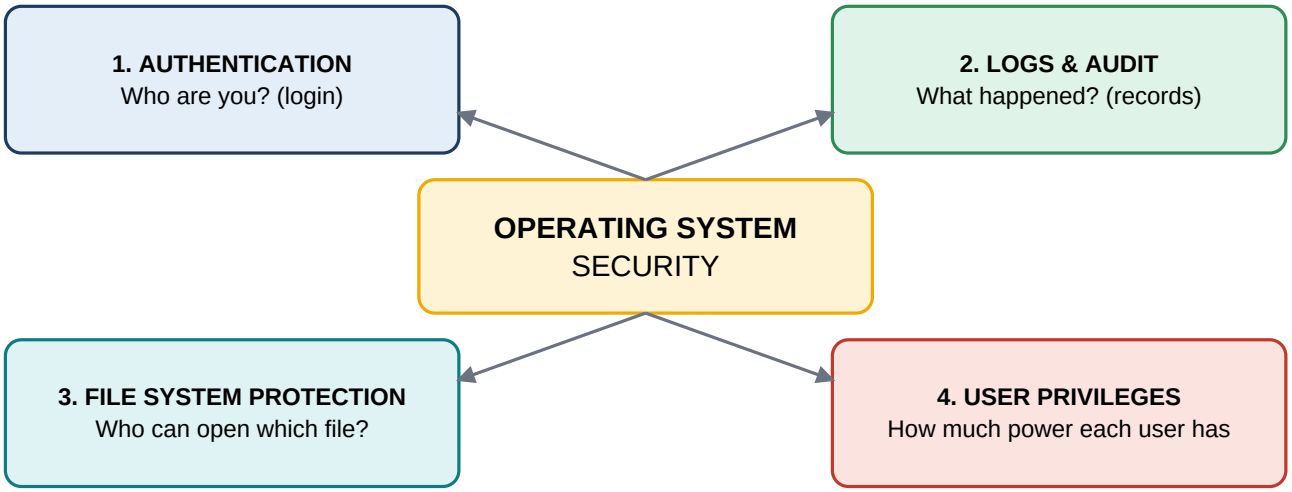


Figure: Security features of an OS

Conclusion: These features work together to protect confidentiality, integrity and availability of data in the operating system.

Q4. What is Anti-virus software? How does it work? VERY LIKELY

Definition / Introduction: Anti-virus software is a program designed to detect, prevent and remove malicious software such as viruses, worms, trojans and ransomware.

Points

- **Signature-based detection:** compares files with a database of known virus signatures.
- **Heuristic detection:** analyses code for suspicious patterns to find unknown viruses.
- **Behaviour-based detection:** monitors programs while running and blocks harmful actions.
- **Sandboxing:** runs suspicious files in an isolated environment to observe them safely.
- **Features:** real-time protection, scheduled scans, quarantine, automatic updates, email and web protection.
- **Limitation:** new (zero-day) threats may escape until definitions are updated.

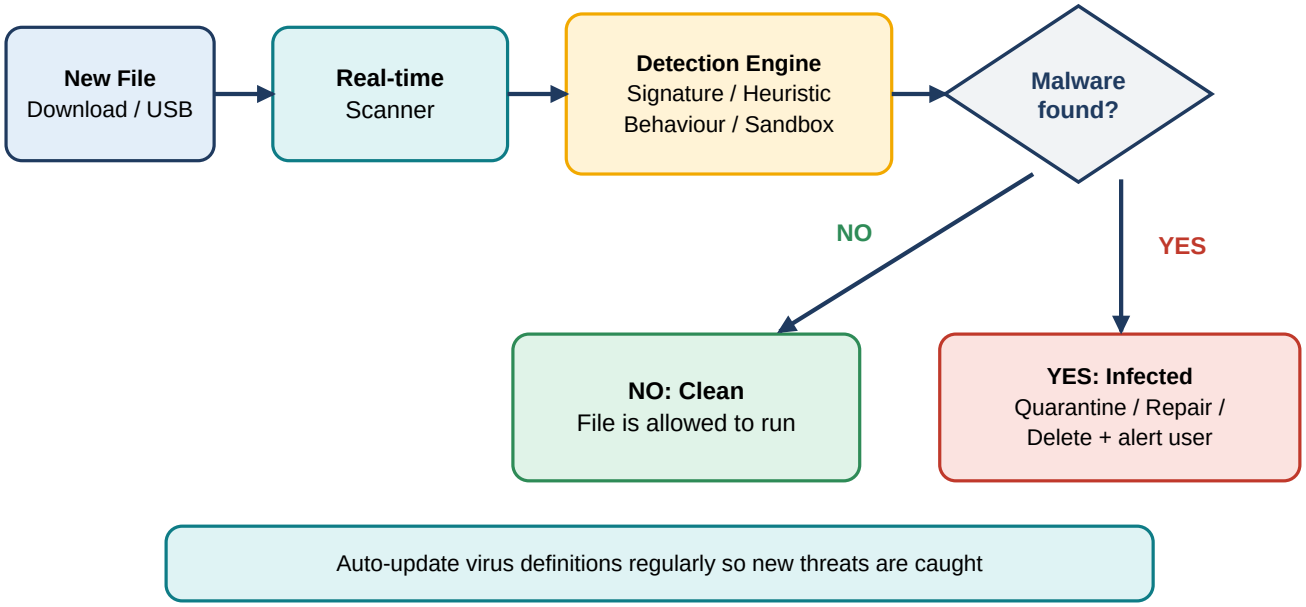


Figure: Working of anti-virus

Conclusion: Anti-virus with regular updates is an essential layer of protection for any computer system.

Q5. What is Authentication? Explain its types. LIKELY

Definition / Introduction: Authentication is the process of verifying the identity of a user, device or process before granting access to a system.

Points

- **Knowledge factor (something you know):** password, PIN, security question.
- **Possession factor (something you have):** OTP, smart card, hardware token, mobile phone.
- **Inherence factor (something you are):** fingerprint, face, iris or voice recognition.
- Using two or more factors is called **Multi-Factor Authentication (MFA)**. Example: ATM card plus PIN.

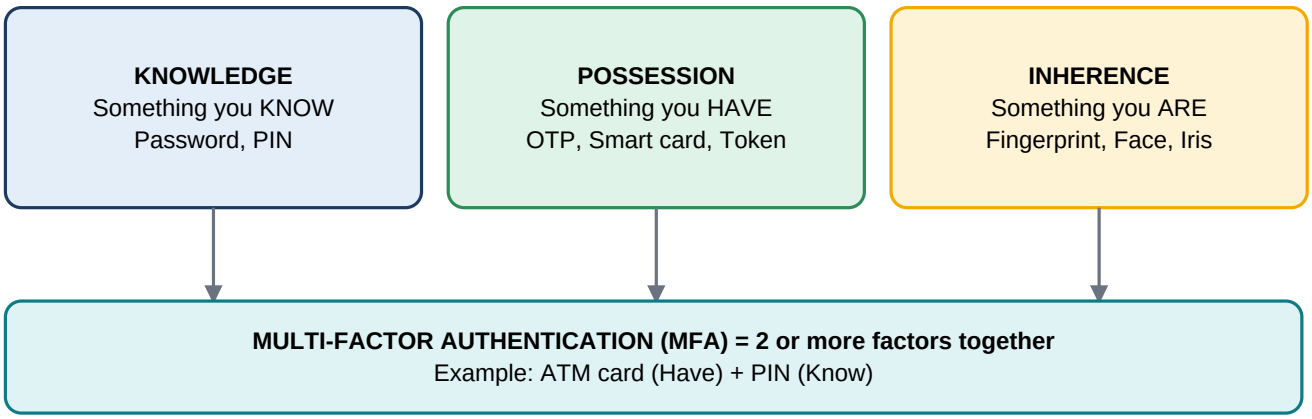


Figure: Authentication factors

Authentication	Authorization
Verifies **who** the user is	Decides **what** the user is allowed to do
Happens first (login)	Happens after authentication
Example: entering password	Example: user can read but not delete a file

Conclusion: Strong authentication, especially MFA, greatly reduces the risk of unauthorized access.

Q6. Explain Logs and Audit features of an Operating System. LIKELY

Definition / Introduction: A **log** is a record of events that occur in a system. An **audit** is the process of reviewing these logs to check for security violations and unusual activity.

Points

- Logs record user logins, failed login attempts, file access, software installation and system errors, along with date, time and user.
- Audit helps to detect attacks, find the source of a breach and prove accountability.
- Logs also help in troubleshooting and in meeting legal and compliance requirements.
- **Windows** stores logs in **Event Viewer**. **Linux** stores logs in the `/var/log/` directory.
- Logs must be protected from modification, otherwise an attacker can erase his traces.

Log type	What it records
Security log	Logins, logouts, failed attempts, permission changes
System log	OS events such as startup, shutdown, driver errors
Application log	Events and errors from installed programs

Conclusion: Logging and auditing provide visibility and accountability, which are essential for detecting and investigating security incidents.

Q7. Explain File System Protection. LIKELY

Definition / Introduction: File system protection means controlling which users can read, write or execute files and folders, to prevent unauthorized access or modification.

Points

- **Permissions:** read (r), write (w), execute (x) are assigned separately to Owner, Group and Others in Linux.
- **ACL (Access Control List):** in Windows NTFS, a list that defines which user or group has which rights on an object.
- **Encryption:** EFS encrypts individual files and BitLocker encrypts a complete drive.
- **Other methods:** passwords on files, hidden or read-only attributes, regular backups.

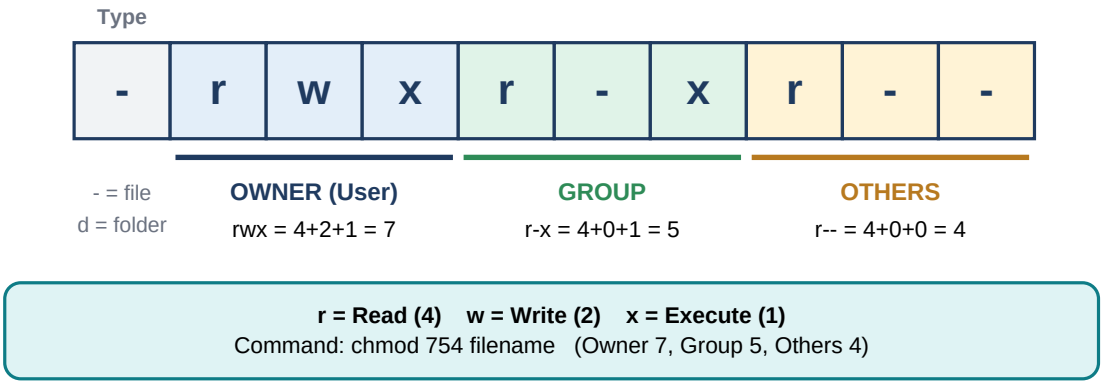


Figure: Linux permission -rwxr-xr-- (chmod 754)

Conclusion: File system protection keeps data confidential and unmodified by allowing only authorized access.

Q8. What are User Privileges? Explain the Principle of Least Privilege. LIKELY

Definition / Introduction: User privileges are the rights and permissions assigned to a user account, which decide what actions the user can perform on the system.

Points

- **Principle of Least Privilege:** every user or program should get only the minimum access needed to do its task.
- It limits damage from mistakes, malware and insider misuse.
- **Windows UAC** asks for confirmation before any change needing administrator rights.
- **Linux sudo** gives temporary administrator rights to normal users.

Account type	Privileges
Administrator / root	Full control: install software, change settings, manage users
Standard user	Run programs and use own files, limited system changes
Guest	Very limited, temporary access

Conclusion: Proper privilege management reduces the attack surface and improves overall system security.

Q9. Explain the CIA Triad. LIKELY

Definition / Introduction: The CIA Triad is a model with three core goals of information security: Confidentiality, Integrity and Availability.

Points

- **Confidentiality:** only authorized users can access data. Achieved by encryption, passwords, access control.
- **Integrity:** data remains accurate and unchanged. Achieved by hashing, checksums, digital signatures.
- **Availability:** data and systems are accessible when needed. Achieved by backups, RAID, redundancy, anti-DoS measures.

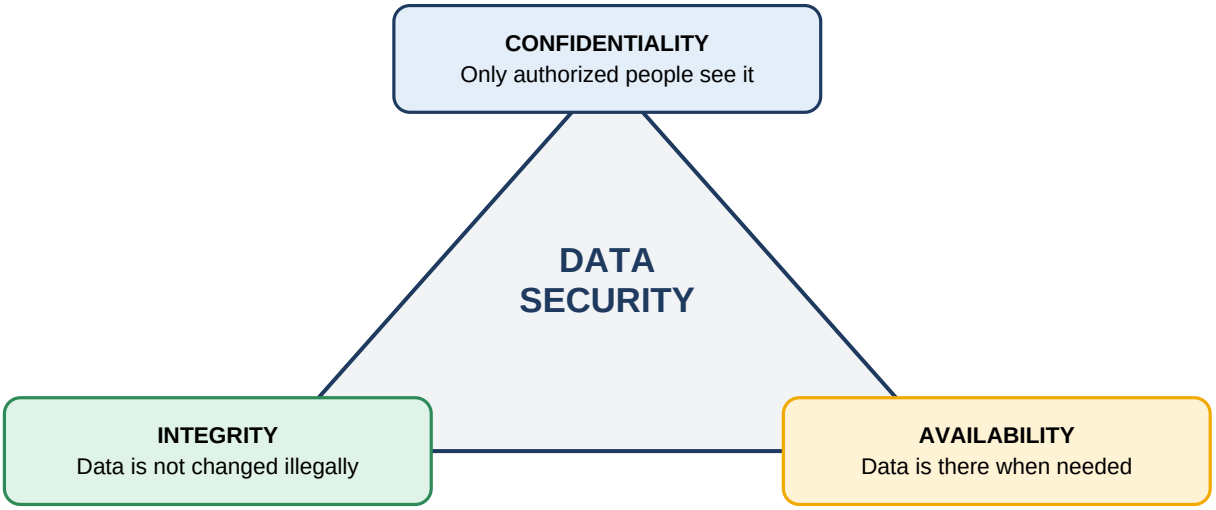


Figure: CIA Triad

Conclusion: A secure system must balance all three goals of the CIA Triad.

Q10. Differentiate between RAID 0 and RAID 1. POSSIBLE

Definition / Introduction: RAID 0 uses striping to increase speed, whereas RAID 1 uses mirroring to protect data.

Points

- Both need a minimum of two disks, but their purpose is different.

Point	RAID 0	RAID 1
Technique	Striping	Mirroring
Main goal	High performance	Data safety
Fault tolerance	None	Survives one disk failure
Usable capacity	100%	50%
Cost	Low	Higher (double disks)
Use case	Video editing, gaming	Important data, servers

Conclusion: RAID 0 is chosen for speed and RAID 1 for reliability.

Q11. Differentiate between Signature-based and Heuristic-based detection. POSSIBLE

Definition / Introduction: Both are methods used by anti-virus software to identify malware.

Points

- Signature-based detection looks for known patterns, while heuristic detection looks for suspicious behaviour or code.

Point	Signature-based	Heuristic-based
Method	Matches file with database of known virus signatures	Analyses code/behaviour for suspicious traits
Detects	Only known viruses	Known and unknown (new) viruses
Accuracy	Very accurate, few false alarms	May give false positives
Update need	Needs frequent database updates	Less dependent on updates

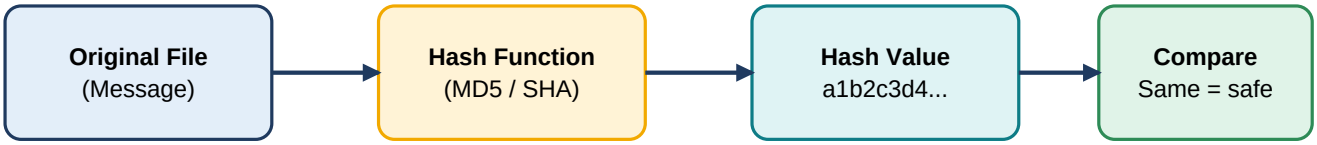
Conclusion: Modern anti-virus combines both methods for better protection.

Q12. What is Non-repudiation? Explain with an example. POSSIBLE

Definition / Introduction: Non-repudiation ensures that the sender of a message or the performer of a transaction cannot later deny having done it.

Points

- It is mainly provided by **digital signatures**, which are created using the sender's private key and verified with the public key.
- It also depends on audit logs and timestamps as proof.
- **Example:** if a customer approves an online payment using OTP and digital proof, he cannot later claim that he never made the payment.
- It is important in e-commerce, banking and legal contracts.



If even 1 character changes, the hash value changes completely (Integrity check)

Figure: Hash value used to prove integrity of a signed message

Conclusion: Non-repudiation gives legal proof of actions and builds trust in digital transactions.

Quick Revision Sheet (read the night before)

Term	Remember this
Information Security	Protect data from unauthorized access, change, destruction
CIA	Confidentiality, Integrity, Availability
PAIN	Privacy, Authentication, Integrity, Non-repudiation
Authentication factors	Know (password), Have (OTP, card), Are (fingerprint)
Authentication vs Authorization	Who are you? vs What can you do?
Log / Audit	Log = record of events. Audit = checking the logs
Windows / Linux logs	Event Viewer / `/var/log/`
Linux permissions	r=4, w=2, x=1. Owner, Group, Others. chmod 754
Windows file security	NTFS ACL, EFS (file), BitLocker (drive)
Least privilege	Give only the minimum access needed
UAC / sudo	Windows admin prompt / Linux temporary admin
RAID full form	Redundant Array of Independent Disks
RAID 0 / 1 / 5	Striping, no safety / Mirror, min 2 disks / Parity, min 3 disks
RAID is not backup	Deleted file or virus is copied to all disks
Anti-virus methods	Signature, Heuristic, Behaviour, Sandbox
Quarantine	Isolating an infected file safely
Zero-day	Brand new threat not yet in virus database

Memory tricks

RAID numbers: 0 = Zero safety, 1 = One copy (mirror), 5 = Parity (minimum **3** disks).
Authentication factors: K-H-A = Know, Have, Are.
PAIN: P-A-I-N, the same word as 'pain', which you feel if you ignore security.
Anti-virus methods: S-H-B-S = Signature, Heuristic, Behaviour, Sandbox.

Common Mistakes Students Make

- Writing **Authorization** instead of **Authentication** in the PAIN full form. The correct word is Authentication.
- Mixing Authentication (who are you?) and Authorization (what can you do?).
- Calling RAID a backup. It is not.
- Saying RAID 0 is safe. It has **no** redundancy.
- Not drawing diagrams. In a 5-mark question, a neat labelled diagram secures marks.
- Writing only the definition. Always add points, an example and a conclusion.
- Forgetting examples such as ATM card + PIN for MFA, Windows Defender for anti-virus.

Self-Test (answer without looking, 5 marks each)

- Write the 4 aspects of PAIN with one example of each.
- Draw and explain RAID 0, RAID 1 and RAID 5. Give a comparison table.
- What are the three factors of authentication? What is MFA?
- Differentiate between Authentication and Authorization.
- How does signature-based detection work in anti-virus software? What is its limitation?

Self-Test Answers (check after writing)

1. Privacy (WhatsApp end-to-end encryption), Authentication (ATM PIN), Integrity (hash match after file download), Non-repudiation (digital signature on an online payment).

2. RAID 0: striping, min 2 disks, no fault tolerance. RAID 1: mirroring, min 2 disks, survives 1 failure, 50% capacity. RAID 5: striping with distributed parity, min 3 disks, survives 1 failure. Draw Figure 6 and the comparison table from Q2.

3. Something you know (password, PIN), something you have (OTP, smart card), something you are (fingerprint, face). MFA means using two or more of these together, for example ATM card + PIN.

4. Authentication verifies **who** the user is (login). Authorization decides **what** the user can do after login (permissions). Authentication comes first.

5. It compares each file with a database of known virus signatures (unique patterns). If it matches, the file is quarantined or deleted. Limitation: it cannot detect new (zero-day) viruses until the signature database is updated.

> **Done with Unit 1!**
> If you scored 4 out of 5 or more, move to **Unit 2**. If not, re-read the topics you missed in Part A.